

Three-Dimensional Free Vibration Analysis and Critical Speed of Pressurized Rotating Functionally Graded Cylindrical Shells

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Abstract This paper presents an approximate solution for the free vibration analysis of rotating functionally graded cylindrical shells based on the three-dimensional theory, using layerwise theory. Equations of motion are derived by applying Hamilton's principle. In order to accurately account for the thickness effects, the layerwise theory is used to discretize the equations of motion and the related boundary conditions through the thickness of the shells. The edges of the shell are restrained by simply supported boundary conditions. Material properties are assumed graded in the thickness direction according to a simple power-law distribution in terms of the volume fraction of the constituents. The results obtained include the relationship between frequency characteristics, different material properties, power-law index, rotating velocities and amplitude of internal pressure. In order to validate the present analysis, the comparison is made with the other published works for a cylindrical shell obtained from CLT to TSDT. In addition, another comparison is made with non-rotating isotropic cylindrical shell based on 3D theory. These comparisons confirm reliability of the present work as a measure to approximate solutions to the problem of rotating functionally graded cylindrical shells.

Keywords Rotating cylindrical shell · Functionally graded material · Three-dimensional elasticity theory · Critical speed · Natural frequency

1 Introduction

Circular shell structures are increasingly being used in many engineering applications. Vibration of shell structure has been extensively studied (Leissa 1973; Soedel 1996; Reddy 2004; Najafi and Ahmadzadeh 2014). Rotating cylindrical shells are being widely used in the drive shafts of gas turbines, high-speed centrifugal separators, motors and rotor systems.

Although rotating shells of revolution are being more frequently used in many industries, most studies were restricted to the vibration analysis of rotating cylindrical shells on basis 2D theory. They include the works by Lam on the rotating composite and sandwich-type cylindrical shells (Lam and Loy 1994, 1995) as well as a comparative study on different thin shell theories and also a discussion on influence of boundary condition for rotating cylindrical shell (Lam and Loy 1995; Hua and Lam 1998). The vibration and critical speed of thin, isotropic cylindrical shells under constant axial loads is studied by Ng and Lam (1999). Daneshjou et al. (2013) employed the generalized differential quadrature method (GDQM) to consider the free vibration and critical speed of moderately thick rotating laminated composite conical shells with different boundary conditions developed from the first-order shear deformation theory (FSDT). Free vibration of orthotropic and cross-ply laminated hemispherical shells using 3D elasticity approach is studied by Shakeri and Yas (2002). The partial differential equations are reduced to ordinary differential equations, by applying Galerkin method, using Legendre polynomials along the meridian direction.

The functionally graded material (FGM) is a new kind of composite material made from a mixture of metals and ceramics and is further characterized by a smooth and continuous change of the mechanical properties from one

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surface to another (Sofiyev 2009). The original purpose of FGMs was the development of super resistant materials for high-temperature environments in such a manner that ceramic can resist the severe thermal loading, and the metal is served to decrease the large tensile stress occurring on the ceramic surface at the earlier stage of cooling. Several works on FGM cylindrical shells have been carried out. Loy et al. (1999) obtained the natural frequencies of simply supported FGM cylindrical shells applying Love's shell theory and Rayleigh–Ritz method. By applying the same method, Pradhan et al. (2000) studied the vibration characteristics of FGM cylindrical shells under various boundary conditions. The buckling analysis of radially loaded solid circular plate made of functionally graded material is investigated by Najafizadeh and Eslami (2002). In this research, the edge of the plate is either simply supported or clamped and the plate is assumed to be geometrically perfect. Najafizadeh and Isvandzibaei (2007) studied the vibration of thin cylindrical shells with ring supports made of a functionally gradient material (FGM) composed of stainless steel and nickel based on TSDT. Closed-form solution for the elastic buckling of stiffened cylindrical shells stiffened by rings and stringers made of functionally graded materials subjected to axial compression loading is carried out by Najafizadeh et al. (2009). The influences of ring support position and boundary conditions on the frequency characteristics of thin FG cylindrical shells using various shear deformation theories are studied by Najafizadeh and Isvandzibaei (2009). Malekzadeh and Heydarpour (2012) analyzed the free vibration analysis of rotating functionally graded (FG) cylindrical shells subjected to thermal environment based on the FSDT using differential quadrature method. With the aid of generalized differential-integral quadrature method (GDIQM), the free vibration of a two-dimensional functionally graded circular cylindrical shell is analyzed by Ebrahimi and Najafizadeh (2013). Malekzadeh et al. (2013) studied the free vibration analysis of functionally graded (FG) cylindrical panels with a cut-out and under thermal environment using the three-dimensional Chebyshev–Ritz method. The material properties of panels under consideration were assumed to be temperature-dependent and graded in the thickness direction. The dynamic behavior of moderately thick functionally graded cylindrical shell surrounded by Winkler–Pasternak elastic foundation based on the FSDT is investigated by Bahadori and Najafizadeh (2015). Navier-differential quadrature solution method is used to obtain natural frequencies and mode shapes. Recently, Nami and Janghorban (2015) developed an analytical solution to study the free vibration analysis of functionally graded rectangular nanoplates. The governing equations of motion are derived based on second-order shear deformation theory using non-local elasticity theory. It is assumed that the

material properties of nanoplate vary through the thickness according to the power-law distribution.

To the best knowledge of the authors, there is no work available in the literature on the subject of critical speed of pressurized FG cylindrical shells by the means of 3D theory of elasticity. Previous works on vibration analysis of 2D FG cylindrical shells have been done using CLT or HSDT. The main advantage of the suggested fast converging method in this paper is conserving reliability of the results for thick FG cylindrical shells while reducing the amount of computations. In this paper, a theoretical analysis is presented to study the critical speed of rotating FG cylindrical shells under internal pressure. The analysis is carried out using layerwise approach for thick shells together on the basis of 3D elasticity theory with consideration of the initial hoop tension and centrifugal and Coriolis forces. The properties are graded in the thickness direction according to a volume fraction power-law distribution. Numerical results are presented for different values of power-law exponent, circumferential wave number, effects of internal pressure and various length and thickness to radius ratios of the cylindrical shells.

2 Problem Formulation

2.1 Geometrical Configuration

The cylindrical shell, as shown in Fig. 1, is considered to be thick, and its mechanical properties vary continuously due to a gradual change in the volume fraction. In this figure, L is the length, h is the thickness, and Ω is the constant angular velocity of cylindrical shell about its symmetrical and horizontal axis. Reference surface of the cylindrical shell is taken to be at its middle surface where an orthogonal coordinate system (x, θ, z) is fixed, and R_j is radius of the j th layer. In the layerwise approach, the cylinder is discretized into an arbitrary number of thin annular cylindrical layers and displacement of the shell in x , θ and z directions is denoted by u^j , v^j and w^j , respectively.

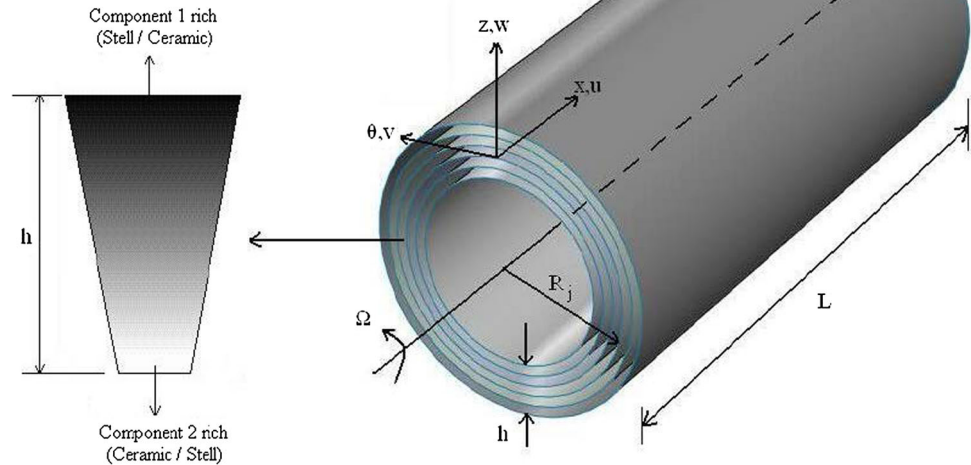
2.2 Functionally Graded Materials

In a FG material made of ceramic and metal mixture, with the material composition varying smoothly along its thickness direction, the material properties of FGMs F_{FGM} , like Young's modulus E_{FGM} or Poisson's ratio ν_{FGM} or density ρ_{FGM} , can be expressed as

$$F_{\text{FGM}} = F_c V_c + F_m V_m \quad (1)$$

where F_c and F_m are the material properties of the ceramic and metal, respectively.

Fig. 1 Geometry of a rotating functionally graded cylindrical shell with coordinate convention



If the volume fraction of the ceramic part is represented by V_c and the metallic part by V_m , it can be followed as

$$V_m + V_c = 1 \quad (2)$$

In this study, the properties of the shell are assumed to vary through the thickness of the shell with a power-law distribution of the volume fraction of the two materials between the two surfaces. In fact, the top surface of the shell is metal rich, whereas the bottom surface is ceramic rich.

Based on the power-law distribution (Ebrahimi and Rastgo 2008), the variation of V_c versus thickness coordinate (z) placed at the reference surface can be expressed as

$$V_c(z) = \left(\frac{2z+h}{2h} \right)^g, \quad g \geq 0 \quad (3)$$

where g is the power-law index.

From Eqs. (1) and (2), the effective Young's modulus, mass density and Poisson's ratio of FGMs can be written as

$$\begin{aligned} E(z) &= (E_m - E_c)V_c(z) + E_c; \\ \rho(z) &= (\rho_m - \rho_c)V_c(z) + \rho_c; \\ \nu(z) &= (\nu_m - \nu_c)V_c(z) + \nu_c \end{aligned} \quad (4)$$

where the subscripts m and c represent the metallic and ceramic constituents, respectively.

In order to clarify the behavior of Eq. (4), the variation of Young's modulus in thickness direction with various values of power-law index g is shown in Fig. 2. From Eq. (4), it can be deduced that

$$\begin{aligned} E(z) &= E_c, \quad \rho(z) = \rho_c, \quad \nu(z) = \nu_c \quad \text{at } z = -h/2 \\ E(z) &= E_m, \quad \rho(z) = \rho_m, \quad \nu(z) = \nu_m \quad \text{at } z = h/2 \end{aligned} \quad (5)$$

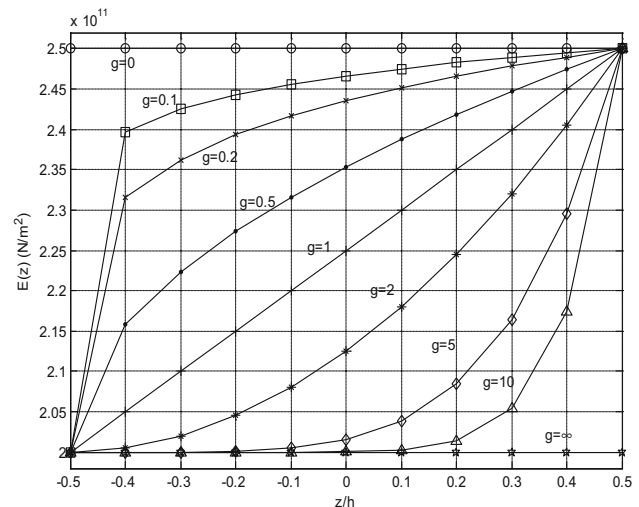


Fig. 2 Variation of Young's modulus in thickness direction with various values of g

2.3 The 3D Theory

According to the three-dimensional theory, stress-strain relation is given by Hooke's law as

$$\{\sigma\} = [C]\{\varepsilon\} \quad (6)$$

where

$$\{\sigma\}^T = \{\sigma_x \sigma_\theta \sigma_z \tau_{\theta z} \tau_{xz} \tau_{x\theta}\}; \quad \{\varepsilon\}^T = \{\varepsilon_x \varepsilon_\theta \varepsilon_z \gamma_{\theta z} \gamma_{xz} \gamma_{x\theta}\} \quad (7)$$

Meanwhile, elastic coefficient matrix $[C]$ for FGM is defined as

$$[C] = \begin{bmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ C_{21} & C_{22} & C_{23} & 0 & 0 & 0 \\ C_{31} & C_{32} & C_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{66} \end{bmatrix} \quad (8)$$

where coefficients C_{ij} are defined as

$$\begin{aligned} C_{11} = C_{22} = C_{33} &= \lambda + 2G; \quad C_{12} = C_{13} = C_{23} = \lambda; \\ C_{44} = C_{55} = C_{66} &= G; \\ G &= \frac{E(z)}{2(1+\nu(z))}; \quad \lambda = \frac{\nu(z)E(z)}{(1+\nu(z))(1-2\nu(z))} \end{aligned} \quad (9)$$

Three-dimensional elasticity strain-displacement relations can be written in a matrix form as

$$\{\varepsilon\} = [D]\{u\} \quad (10)$$

where

$$\{u\}^T = \{u \ v \ w\} \quad (11)$$

and

$$[D] = \begin{bmatrix} \frac{\partial}{\partial x} & 0 & 0 \\ 0 & \frac{1}{R_j} \frac{\partial}{\partial \theta} & \frac{1}{R_j} \\ 0 & 0 & \frac{\partial}{\partial z} \\ 0 & \frac{\partial}{\partial z} - \frac{1}{R_j} & \frac{1}{R_j} \frac{\partial}{\partial \theta} \\ \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} \\ \frac{1}{R_j} \frac{\partial}{\partial \theta} & \frac{\partial}{\partial x} & 0 \end{bmatrix} \quad (12)$$

where R_j is the radius of the j th layer.

2.4 Governing Equations

In this study, Hamilton's principle is used to derive equations of motion. For the j th layer subjected to simply supported boundary conditions, the displacements u^j , v^j and w^j are written as

$$\begin{aligned} u^j &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \varphi_x^j(z) \cos \frac{m\pi x}{L} \cos(n\theta + \omega t); \quad v^j \\ w^j &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \varphi_z^j(z) \sin \frac{m\pi x}{L} \cos(n\theta + \omega t) \end{aligned} \quad (13)$$

where

$$\begin{aligned} \varphi_x^j(z) &= \alpha U^j + \beta U^{j+1}; \quad \varphi_\theta^j(z) = \alpha V^j + \beta V^{j+1}; \quad \varphi_z^j(z) \\ &= \alpha W^j + \beta W^{j+1}; \quad \alpha = 1 - \beta; \quad \beta = \frac{z - z_j}{h} \end{aligned} \quad (14)$$

where z_j is the distance from the middle surface of the cylinder to the lower surface of j th layer and U^j , V^j , W^j , U^{j+1} , V^{j+1} and W^{j+1} are the generalized coordinates.

The displacement fields defined in Eq. (13) can be written in a matrix form as

$$\{u^j\} = [N]\{d^j\} \quad (15)$$

where

$$\{d^j\}^T = \{U^j \ V^j \ W^j \ U^{j+1} \ V^{j+1} \ W^{j+1}\} \quad (16)$$

$$[N]^T = \begin{bmatrix} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \alpha \cos \frac{m\pi x}{L} \cos(n\theta + \omega t) & 0 & 0 \\ 0 & \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \alpha \sin \frac{m\pi x}{L} \sin(n\theta + \omega t) & 0 \\ 0 & 0 & \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \alpha \cos \frac{m\pi x}{L} \sin(n\theta + \omega t) \\ \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \beta \cos \frac{m\pi x}{L} \cos(n\theta + \omega t) & 0 & 0 \\ 0 & \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \beta \sin \frac{m\pi x}{L} \sin(n\theta + \omega t) & 0 \\ 0 & 0 & \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \beta \cos \frac{m\pi x}{L} \sin(n\theta + \omega t) \end{bmatrix} \quad (17)$$

and substituting Eq. (15) into Eq. (10), the strain vector can be written as

$$\{\varepsilon\} = [B]\{d\} \quad (18)$$

where

$$[B] = [D][N] \quad (19)$$

The strain energy for the j th layer of cylindrical shell is expressed as

$$U_\varepsilon^j = \frac{1}{2} \iiint_{\text{vol}} \sigma^T \cdot \varepsilon R_j d\theta dx dz \quad (20)$$

It should define the work carried out on the shell due to the centrifugal force generated by rotation. This can be written as

$$U_h^j = \frac{1}{2} \iiint_{\text{vol}} N_\theta \left\{ \left(\frac{1}{R_j} \frac{\partial u^j}{\partial \theta} \right)^2 + \left[\frac{1}{R_j} \left(\frac{\partial v^j}{\partial \theta} + w \right) \right]^2 + \left[\frac{1}{R_j} \left(\frac{\partial w^j}{\partial \theta} - v \right) \right]^2 \right\} R_j d\theta dx dz \quad (21)$$

where N_θ is the initial hoop stress due to centrifugal force which is given by

$$N_\theta = \rho \Omega^2 R_j^2 \quad (22)$$

Besides, the work carried out on the shell due to internal pressure is described as

$$U_P^j = -\frac{P}{2} \iiint_{\text{vol}} \left[\left(\frac{\partial^2 w}{\partial \theta^2} + w \right) w \right] R_j d\theta dx dz \quad (23)$$

In addition, kinetic energy of the rotating cylindrical shell is expressed as

$$T^j = \frac{1}{2} \iiint_{\text{vol}} \rho(z) \vec{V} \cdot \vec{V} R_j d\theta dx dz \quad (24)$$

where $\vec{V}$ is the velocity vector at any point of the shell given by

$$\vec{V} = \dot{\vec{r}} + \Omega \vec{i} \times \vec{r} \quad (25)$$

Here, $(\cdot)$ presents differentiation with respect to time.

In Eq. (25) the displacement vector is written as

$$\vec{r} = u^j \vec{i} + v^j \vec{j} + w^j \vec{k} \quad (26)$$

where $\vec{i}$, $\vec{j}$ and $\vec{k}$, respectively, denote the unit vectors in x , θ and z directions for non-rotating frame.

Having substituted Eqs. (25) into Eq. (24), the kinetic energy expression of the j th layer of the shell can be expanded in the form as follows:

Table 1 Mechanical properties of the material

Material	E (GPa)	ρ (kg/m ³)	ν
Ceramic	250	8900	0.33
Metal	200	7800	0.30

$$T^j = \underbrace{\frac{1}{2} \iiint_{\text{vol}} \rho(z) \left[(\dot{u}^j)^2 + (\dot{v}^j)^2 + (\dot{w}^j)^2 \right] R_j d\theta dx dz}_{T_1^j} + \underbrace{\frac{1}{2} \iiint_{\text{vol}} \rho(z) [2\Omega(v\dot{w} - w\dot{v})] R_j d\theta dx dz}_{T_2^j} + \underbrace{\frac{1}{2} \iiint_{\text{vol}} \rho(z) [\Omega^2(v^2 + w^2)] R_j d\theta dx dz}_{T_3^j} \quad (27)$$

Substituting Eq. (15) into Eq. (27) and also Eqs. (18) and (19) into Eq. (20), the first part of kinetic energy and strain energy for the j th layer can be written in a matrix form as

$$T_1^j = \frac{1}{2} \{\dot{d}\}^T [M^j] \{\dot{d}\} ; \quad U_\varepsilon^j = \frac{1}{2} \{d\}^T [K_1^j] \{d\} \quad (28)$$

where M_j and K_1^j are the mass and stiffness matrices of the j th layer defined as

$$M^j = \frac{\pi L}{2} \int_{-\frac{h}{2}}^{\frac{h}{2}} \begin{bmatrix} \alpha_j^2 & 0 & 0 & \alpha_j \beta_j & 0 & 0 \\ 0 & \alpha_j^2 & 0 & 0 & \alpha_j \beta_j & 0 \\ 0 & 0 & \alpha_j^2 & 0 & 0 & \alpha_j \beta_j \\ \alpha_j \beta_j & 0 & 0 & \beta_j^2 & 0 & 0 \\ 0 & \alpha_j \beta_j & 0 & 0 & \beta_j^2 & 0 \\ 0 & 0 & \alpha_j \beta_j & 0 & 0 & \beta_j^2 \end{bmatrix} R_j dz \quad (29)$$

$$K_1^j = \iiint_{\text{vol}} K_{rs}^1 R_j^j d\theta dx dz \quad (30)$$

where K_{rs}^1 are detailed in “Appendix.”

Substituting Eq. (13) into Eqs. (21), (23) and (27), followed by applying Hamilton's principle and furthermore collecting the coefficient of δU^j , δV^j , δW^j , δU^{j+1} , δV^{j+1} and δW^{j+1} , all the terms U_h^j , U_P^j , T_2^j and T_3^j can be written in a matrix form as

$$U_h^j = \pi \int_{-\frac{h}{2}}^{\frac{h}{2}} \int_0^L \underbrace{\begin{bmatrix} \frac{n^2 N_0^2 A^2}{R_j^2} & 0 & 0 & \frac{n^2 N_0^2 BA}{R_j^2} & 0 & 0 \\ 0 & \frac{(n^2+1)N_0 F^2}{R_j^2} - \rho(z)\Omega^2 F^2 & \frac{2nN_0 F^2}{R_j^2} & 0 & \frac{(n^2+1)N_0 HF}{R_j^2} - \rho(z)\Omega^2 HF & \frac{2nN_0 HF}{R_j^2} \\ 0 & \frac{2nN_0 F^2}{R_j^2} & \frac{(n^2+1)N_0 F^2}{R_j^2} - \rho(z)\Omega^2 F^2 & 0 & \frac{2nN_0 HF}{R_j^2} & \frac{(n^2+1)N_0 HF}{R_j^2} - \rho(z)\Omega^2 HF \\ \frac{n^2 N_0^2 BA}{R_j^2} & 0 & 0 & \frac{n^2 N_0^2 B^2}{R_j^2} & 0 & 0 \\ 0 & \frac{(n^2+1)N_0 HF}{R_j^2} - \rho(z)\Omega^2 HF & \frac{2nN_0 HF}{R_j^2} & 0 & \frac{(n^2+1)N_0 H^2}{R_j^2} - \rho(z)\Omega^2 H^2 & \frac{2nN_0 H^2}{R_j^2} \\ 0 & \frac{2nN_0 HF}{R_j^2} & \frac{(n^2+1)N_0 HF}{R_j^2} - \rho(z)\Omega^2 HF & 0 & \frac{2nN_0 H^2}{R_j^2} & \frac{(n^2+1)N_0 H^2}{R_j^2} - \rho(z)\Omega^2 H^2 \end{bmatrix}}_{K_2^j} dx dz \quad (31)$$

$$U_P^j = \pi \int_{-\frac{h}{2}}^{\frac{h}{2}} \int_0^L \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & Pn^2 A^2 & 0 & 0 & Pn^2 BA \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & Pn^2 BA & 0 & 0 & Pn^2 B^2 \end{bmatrix}}_{K_3^j} dx dz \quad (32)$$

$$T_2^j = \frac{\pi L}{2} \int_{-\frac{h}{2}}^{\frac{h}{2}} \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -2\rho(z)\Omega x_j^2 & 0 & 0 & -2\rho(z)\Omega \beta_j x_j \\ 0 & -2\rho(z)\Omega x_j^2 & 0 & 0 & -2\rho(z)\Omega \beta_j x_j & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -2\rho(z)\Omega \beta_j x_j & 0 & 0 & -2\rho(z)\Omega \beta_j^2 \\ 0 & -2\rho(z)\Omega \beta_j x_j & 0 & 0 & -2\rho(z)\Omega \beta_j^2 & 0 \end{bmatrix}}_{G^j} dz \quad (33)$$

$$T_3^j = \pi \int_{-\frac{h}{2}}^{\frac{h}{2}} \int_0^L \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\rho(z)\Omega^2 F^2 & 0 & 0 & -\rho(z)\Omega^2 HF & 0 \\ 0 & 0 & -\rho(z)\Omega^2 F^2 & 0 & 0 & -\rho(z)\Omega^2 HF \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\rho(z)\Omega^2 HF & 0 & 0 & -\rho(z)\Omega^2 H^2 & 0 \\ 0 & 0 & -\rho(z)\Omega^2 HF & 0 & 0 & -\rho(z)\Omega^2 H^2 \end{bmatrix}}_{K_4^j} dx dz \quad (34)$$

where

$$\begin{aligned} A &= \alpha_j \cos\left(\frac{m\pi x}{L}\right); & B &= \beta_j \cos\left(\frac{m\pi x}{L}\right); \\ F &= \alpha_j \sin\left(\frac{m\pi x}{L}\right); & H &= \beta_j \sin\left(\frac{m\pi x}{L}\right) \end{aligned} \quad (35)$$

Kinetic and potential energies of the shell are obtained, adding kinetic and potential energies of all the layers together. In order to arrange the eigenvalue equation according to ω , it is written in the following matrix form,

$$\{ [M]\omega^2 + [G]\omega + [K] \} \{ \delta \} = 0 \quad (36)$$

where

$$\begin{aligned} [K] &= [K_1 + K_2 + K_3 + K_4]; \{ \delta \}^T \\ &= \{ U^1 \ V^1 \ W^1 \ U^2 \ V^2 \ W^2 \dots U^{n_l+1} \ V^{n_l+1} \ W^{n_l+1} \} \end{aligned} \quad (37)$$

Equation (36) is a nonstandard eigenvalue equation. For a given frequency, it can be equivalently transformed into a standard form of eigenvalue equation as (Dong 1997),

$$\left(\begin{bmatrix} 0 & I \\ -K & -G \end{bmatrix} - \begin{bmatrix} I & 0 \\ 0 & M \end{bmatrix} \omega \right) \begin{Bmatrix} \delta \\ \omega \delta \end{Bmatrix} = 0 \quad (38)$$

where I is the $(N_l + 1) \times (N_l + 1)$ identity matrix where N_l represents the number of mathematical layers in the thickness direction.

Using a conventional eigenvalue approach, the standard eigenvalue Eq. (38) can be solved and $2(N_l + 1)$ eigenvalues ω_i are obtained. From these eigenvalues, the two eigenvalues are chosen for which the absolute of real values are the smallest. One of these eigenvalues is negative and corresponds to the backward wave, and the other one is positive and corresponds to the forward wave.

3 Results and Discussion

In the presentation of results shown by figures, the backward and forward waves are presented as a solid line and dashed line, respectively, the unit of rotating speed Ω is Rev/s (revolutions per second or Hz). Material properties of the shells used in this study are given in Table 1.

In order to confirm the validity of the present analysis, the results are compared with available published results in the literature. Firstly, the power-law index of FGMs goes into infinity and the rotating speed of the shell becomes zero in order to model non-rotating isotropic metal cylinder. In Table 2, a comparison is made between the theoretical frequency parameter $\bar{\omega} = \omega L \sqrt{\rho(1+\nu)/E}$ of non-

Table 2 Comparison of frequency parameters $\bar{\omega}$ for a isotropic cylinders ($m = 1$, $\nu = 0.3$)

h/R	$\bar{\omega} = \omega L \sqrt{\rho(1 + \nu)/E}$		Present	h/L	$\bar{\omega} = (\omega h/\pi) \sqrt{\rho/G}$	
	$L/R = 1$				$n = 2$	$h/R = 0.4$
	n	LOY (Loy and Lam 1999)			LOY (Loy and Lam 1999)	Present
0.1	1	1.0623	1.0637	0.01	0.0621	0.0627
	2	0.8825	0.8829	0.1	0.0745	0.0751
	3	0.8095	0.8101	0.2	0.1249	0.1261
	4	0.8989	0.8996	0.4	0.2719	0.2754
	5	1.1220	1.1255	0.6	0.4415	0.4460
0.2	1	1.1889	1.1896	0.8	0.6254	0.6299
	2	1.1010	1.1025	1	0.8170	0.8211
	3	1.1977	1.1990			
	4	1.4897	1.4951			
	5	1.9140	1.9212			

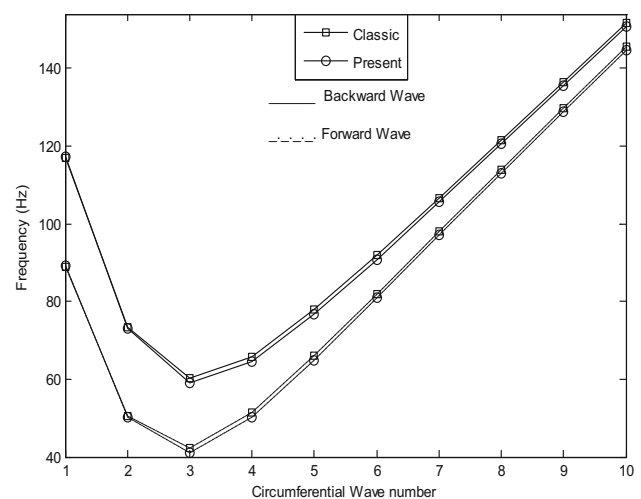
Table 3 Comparison of natural frequencies (Hz) for a non-rotating FG cylindrical shell with simply supported boundary condition ($m = 1$, $h/R = 0.002$, $L/R = 20$)

n	$g = 0.5$		$g = 1$		$g = 5$	
	Najafizadeh (Najafizadeh and Isvandzibaei 2007)	Present	Najafizadeh (Najafizadeh and Isvandzibaei 2007)	Present	Najafizadeh (Najafizadeh and Isvandzibaei 2007)	Present
1	13.320	13.320	13.208	13.208	12.915	12.915
2	4.5147	4.5147	4.475	4.475	4.3763	4.3763
3	4.1900	4.1895	4.1569	4.1569	4.0586	4.0584
4	7.1019	7.1007	7.0441	7.0439	6.8723	6.8717
5	11.343	11.341	11.253	11.252	11.978	11.976
6	16.608	16.604	16.474	16.470	16.073	16.069
7	22.848	22.841	22.659	22.653	22.128	22.121
8	30.053	30.042	29.811	29.804	29.077	29.068
9	38.217	38.205	37.913	37.904	37.982	37.970
10	47.348	47.331	46.968	46.953	46.814	46.811

rotating thick isotropic cylindrical shell based on 3D theory by Loy and Lam (1999) and the theory proposed here.

Secondly, a comparison of natural frequencies (Hz) for a non-rotating FG simply supported cylindrical shell is made with the results of Najafizadeh and Isvandzibaei (Najafizadeh and Isvandzibaei 2007) in Table 3. The shell consists of nickel at the inner surface and stainless steel at the outer surface. It can be seen that the two sets of frequencies are very close to each other, but the present theory estimates more accurate results than those found in (Najafizadeh and Isvandzibaei 2007), since the former employs 3D theory while the latter uses TSDT.

Finally, the results are compared with those of Lam and Hua (1999) for a particular case of rotating thin isotropic cylindrical shell as depicted in Fig. 3. With numerical comparisons listed in Tables 2 and 3, it is evident that the presented results are in good agreement with the data available in the literature, which demonstrates accuracy of the current work.

**Fig. 3** Variation of natural frequencies for the rotating isotropic cylindrical shell versus circumferential wave number in comparison with results of the present study and those given in Lam and Hua (1999). [$L = 4$ m, $R = 1$ m, $h = 6$ mm, $m = 1$, $\Omega = 15$ (Rev/s)]

The layerwise theory is especially suitable for consideration of global characteristics such as free vibration or buckling analyses of thick shells. Table 4 gives the results for various thicknesses when the length and radius of the shell are fixed. The stability and convergence of the numerical values are confirmed with different thicknesses of shell. It is also found that the corresponding frequencies of the first circumferential wave number for rotating and non-rotating thin cylindrical shells are converged quickly

in comparison with those of other circumferential wave numbers. It conforms to expectations that the rate of convergence for the thick shell is slower than that of the thin one.

The convergence behaviors are presented in Table 5 for different values of circumferential wave numbers (n) against the number of layers in the thickness direction (N_l). The results are listed for two different values of the thicknesses. It should also be noted that the fast rate of

Table 4 Convergence behavior of the frequency for a simply supported FG cylindrical shell ($L = 10$, $R = 1$, $m = 1$, $g = 5$)

h (mm)	$n = 1 \quad \Omega = 0$		$n = 7 \quad \Omega = 0$	$n = 1 \quad \Omega = 100$	
	N_l	Freq. (Hz)	Freq. (Hz)	Freq. (Hz) bac. wave	Freq. (Hz) for. wave
2	2	50.7553	24.532	147.6301	46.1224
	6	50.7676	23.9876	147.6410	46.1088
	10	50.7704	23.9543	147.6432	46.1061
	14	50.7715	23.9341	147.6448	46.1043
	18	50.7716	23.9259	147.6443	46.1046
	22	50.7720	23.9247	147.6454	46.1035
	26	50.7719	23.9246	147.6450	46.1039
	30	50.7730	23.9232	147.6455	46.1033
10	2	50.6817	122.7246	147.5642	46.2029
	6	50.7493	119.6433	147.6251	46.1292
	10	50.7628	119.3631	147.6372	46.1146
	14	50.7685	119.2742	147.6424	46.1084
	18	50.7717	119.2328	147.6453	46.1049
	22	50.7738	119.2094	147.6471	46.1027
	26	50.7751	119.1945	147.6484	46.1011
	30	50.7762	119.1843	147.6494	46.1000

Table 5 Convergence behavior of the frequency for simply supported FG cylindrical shell [$L = 10$, $R = 1$, $m = 1$, $g = 2$, $\Omega = 20$ (Rev/s)]

$n = 1$	$h = 10$ (mm)		$n = 3$	$h = 10$ (mm)		$n = 7$	$h = 10$ (mm)	
	N_l	Freq. (Hz) Diff (%)		N_l	Freq. (Hz) Diff (%)		N_l	Freq. (Hz) Diff (%)
2	2	70.0566	2	2	64.6399	2	2	187.6069
6	6	70.1229 0.094638	6	6	64.4399 0.309406	6	6	185.5445 1.09932
10	10	70.1361 0.018824	10	10	64.4224 0.027157	10	10	185.3584 0.100299
14	14	70.1417 0.007984	14	14	64.4169 0.008537	14	14	185.2994 0.03183
18	18	70.1449 0.004562	18	18	64.4144 0.003881	18	18	185.2719 0.014841
22	22	70.1469 0.002851	22	22	64.413 0.002173	22	22	185.2563 0.00842
26	26	70.1482 0.001853	26	26	64.4121 0.001397	26	26	185.2465 0.00529
$n = 1$	$h = 50$ (mm)		$n = 3$	$h = 50$ (mm)		$n = 7$	$h = 50$ (mm)	
	N_l	Freq. (Hz) Diff (%)		N_l	Freq. (Hz) Diff (%)		N_l	Freq. (Hz) Diff (%)
2	2	69.7106	2	2	124.0026	2	2	633.4998
8	8	70.0887 0.542385	8	8	119.9983 3.229206	8	8	608.2531 3.985275
14	14	70.1419 0.075904	14	14	119.6729 0.271171	14	14	606.1382 0.347699
20	20	70.1631 0.030224	20	20	119.5584 0.095677	20	20	605.3892 0.123569
26	26	70.1745 0.016248	26	26	119.5005 0.048428	26	26	605.0085 0.062885
32	32	70.1816 0.010118	32	32	119.4655 0.029289	32	32	604.7787 0.037983

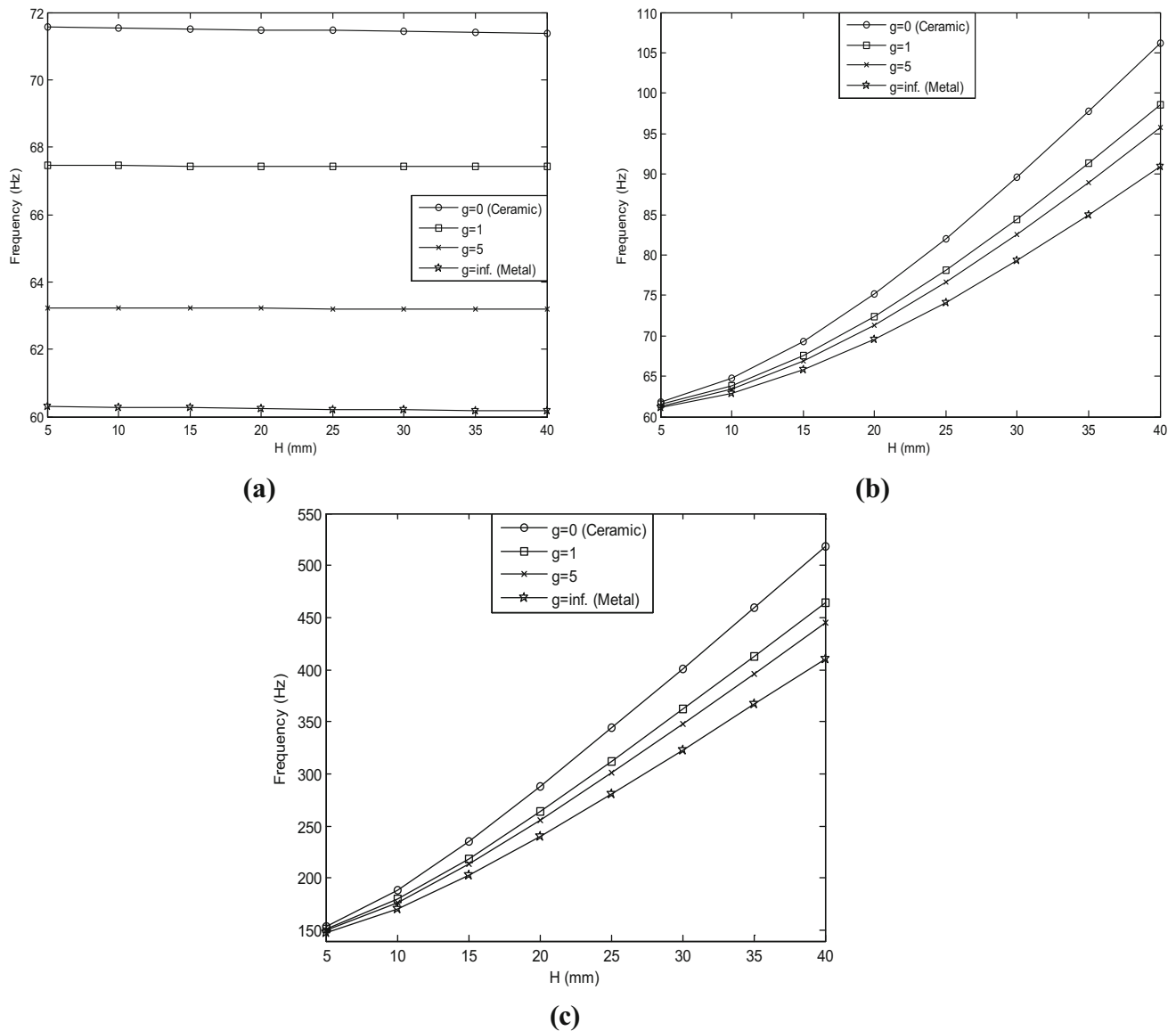


Fig. 4 Variation of frequency with thickness of shell for different power-law index g . **a** $n = 1$, **b** $n = 3$, **c** $n = 7$. [$L = 10$ m, $R = 1$ m, $m = 1$, $\Omega = 20$ (Rev/s), $N_l = 30$]

convergence, the numerical stability and the high accuracy of the method can be seen. Furthermore, monitoring Table 5, it is found that increasing the circumferential wave number and thickness of the shell resulted in the convergence behaviors being delayed.

The effects of the thickness of shell on the frequency of the rotating simply supported FGM shell for various power-law indexes at three circumferential wave numbers are shown in Fig. 4a–c. In this case, rotating speed of shell is 20 (Rev/s). From the figure, it can be observed that at small circumferential wave number (n), i.e., 1, increase in thickness of shell not only increases the frequency but also slightly decreases maximum frequency. This is due to the

fact that the inertial terms of rotating shell are more considerable than the stiffness terms when n is 1. But, at large circumferential wave numbers with the increase in the thickness, frequency increases in a monotonic manner.

As it is well anticipated, increasing the power-law index from zero (ceramics) to infinity (metal), leads to decrease in natural frequency as a result of pushing the stiffness down; this discrepancy behaves the same in all thicknesses where the $n = 1$. However, in upper circumferential modes it appears to be more sensible in higher thicknesses.

Figure 5a–c shows the variation of natural frequency of the backward wave with length of shell for $\Omega = 20$ (Rev/s) and $h = 20$ mm. It is seen that the natural frequencies

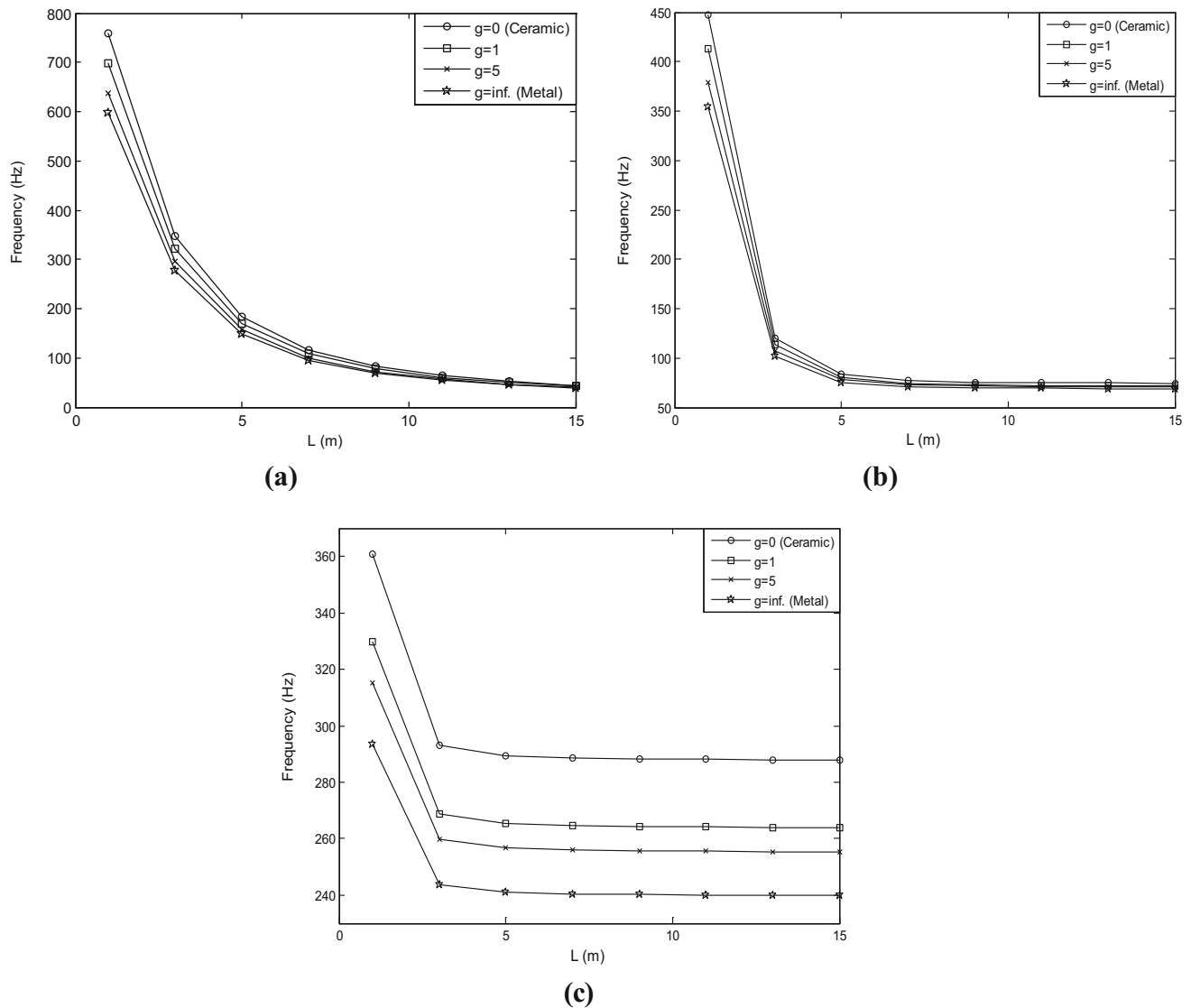


Fig. 5 Variation of frequency with length of shell for different power-law index g . **a** $n = 1$, **b** $n = 3$, **c** $n = 7$. [$R = 1$ m, $h = 20$ mm, $m = 1$, $\Omega = 20$ (Rev/s), $N_1 = 30$]

decrease with increasing length of shell. Also, as the length of the shell increases, natural frequencies of the shells with different circumferential wave numbers converge to the same value. It should be noted that the effects of power-law index (g) on natural frequencies of long shells in lower modes can be negligible. However, it significantly affects on natural frequencies of the shell with short length in a broad band of mode shapes.

In Fig. 6 the natural frequencies of the different shells in thicknesses and lengths versus circumferential modes are depicted in a variety of power-law index (g). As shown, the effects of increasing the thickness of the shell on those natural frequencies corresponding to higher circumferential modes are significant, and the effects appear more sensible

when the number of circumferential modes goes up. On the contrary, these effects are negligible at zero circumferential modes.

Figure 7 plots the natural frequency of the shell for different rotating velocities. The results indicate that increasing the rotating velocity of the shell has an increasing influence on natural frequencies in a broadband frequency. It also causes the fundamental frequency to occur in lower circumferential modes.

The phenomenon of the critical speed of the shell is illustrated in Fig. 8. The critical speed of the rotating shell corresponds to the rotational speed of the shell at which the forward wave intersects the abscissa. At this intersection, possible unstable phenomenon exists as the forward wave

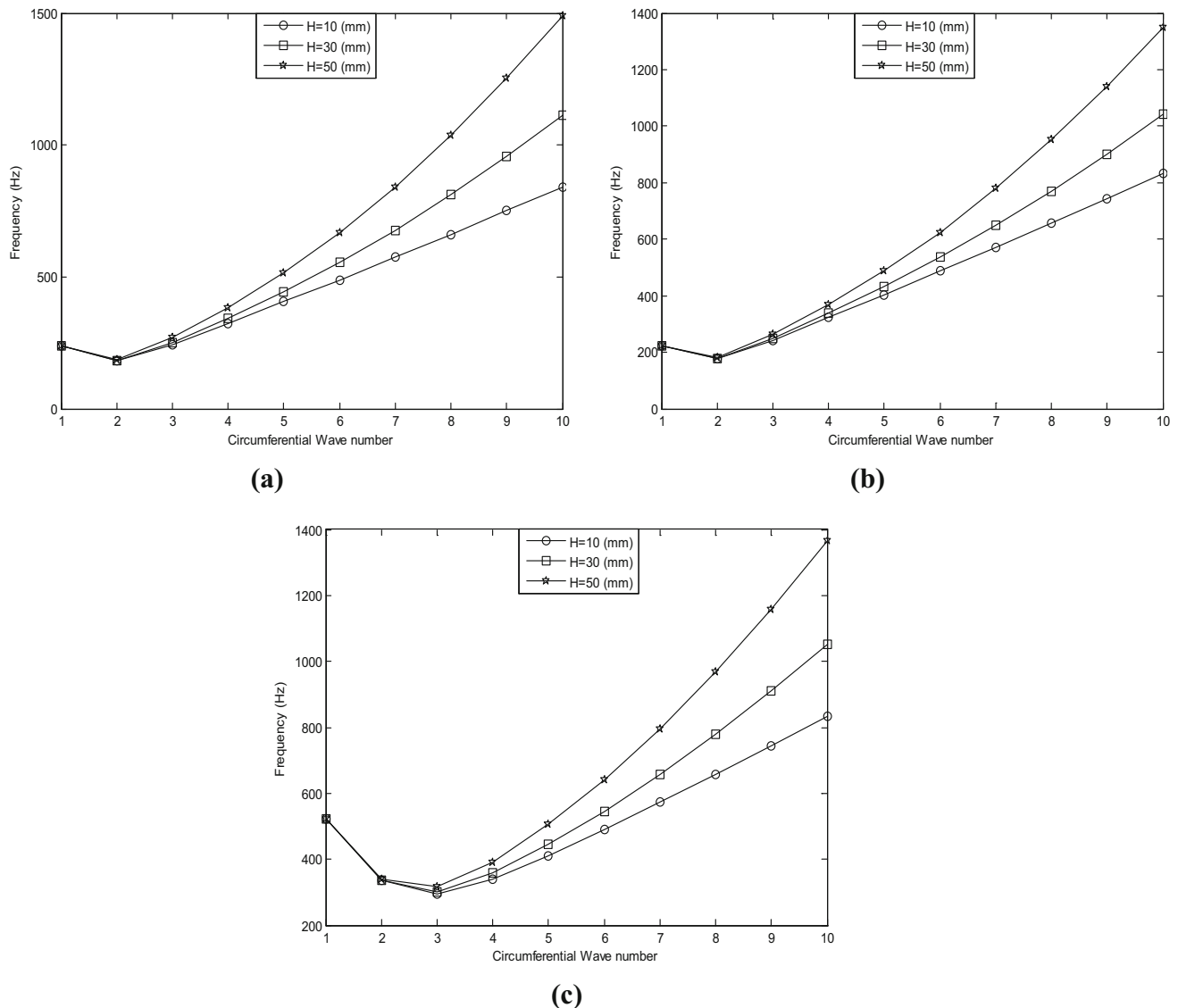


Fig. 6 Variation of frequency with circumferential wave number for different thicknesses of shell. **a** $g = 0$, $L = 5$ m, **b** $g = 2$, $L = 5$ m, **c** $g = 2$, $L = 2$ m [$R = 1$ m, $h = 50$ mm, $m = 1$, $\Omega = 80$ (Rev/s), $N_1 = 30$]

becomes standing with respect to the traveling coordinate and is ready to switch to a backward mode. At this critical speed, any residual unbalance will synchronize with the rotation and magnify the whirling amplitude. It is clearly obvious that this phenomenon in all structures with different materials has occurred in first circumferential modes where it behaves linearly, contrary to all other modes. The increasing influence of power-law index (g) in modes with higher circumferential wave numbers can also be seen.

The critical speed of the pressurized shell for different internal pressures is depicted in Fig. 9. As expected, while the internal pressure increases the natural frequency and

critical speed of the shell will be enhanced as a result of stiffening of the shell. However, the sensitivity of the critical speed in higher pressures will be decreased.

4 Conclusion

Three-dimensional theory of elasticity has been used in this paper to study free vibration and critical speed of the rotating FG cylindrical shells. The effects of different parameters such as geometry, volume fraction and internal pressures on the natural frequency and critical speed of the

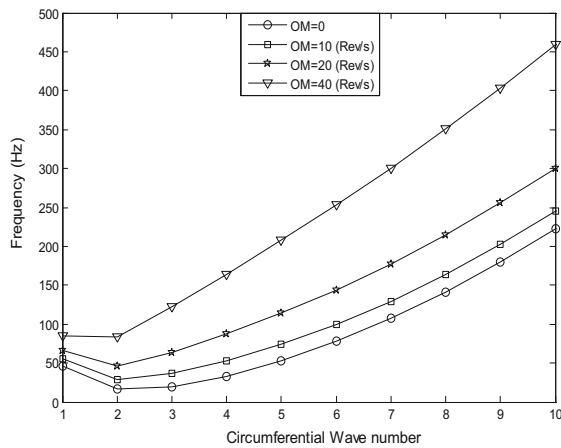
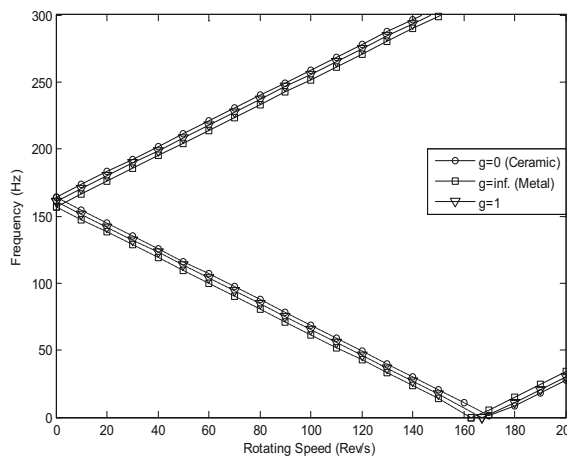
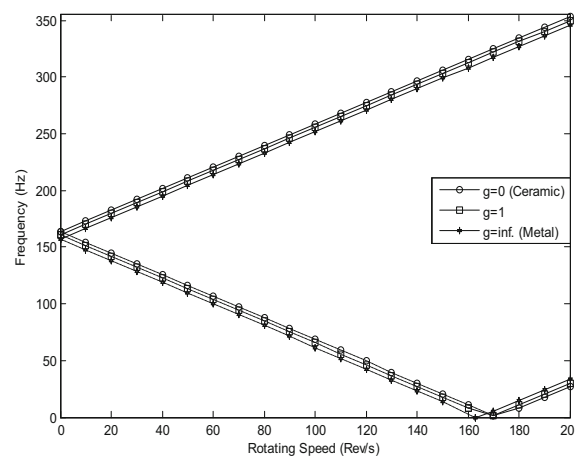


Fig. 7 Variation of frequency with circumferential wave number for different rotating speed Ω . **a** $h = 10$ mm, **b** $h = 50$ mm. ($L = 10$ m, $R = 1$ m, $m = 1$, $g = 2$, $N_1 = 30$)

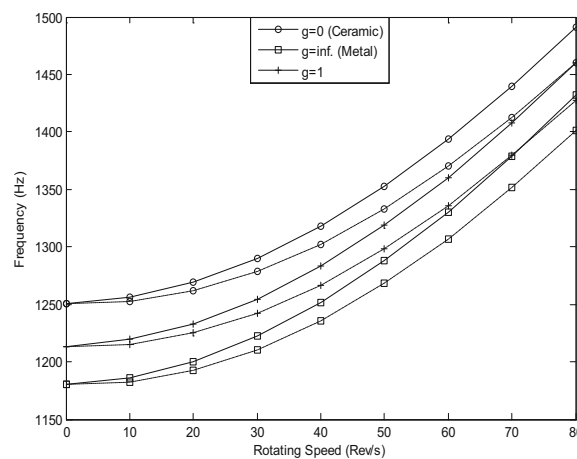
shell are investigated. The results confirm that the layer-wise theory has high reliability and the efficiency of high convergence. In addition, the results indicate that the variation of parameters such as thickness, rotating speed and volume fraction of the shell, in high circumferential wave numbers, is of high sensitivity. Also, the effect arisen from thickness of the shell on the natural frequency at the first circumferential wave number is negligible. The main point that should be considered here is that the critical speed of the shell occurs at the first circumferential wave number disregarding of the other parameters such as axial wave number and internal pressure. Finally, the natural frequency and critical speed are enhanced due to the increase in the internal pressure of the shell, but its sensitivity will be decreased.



(a)



(b)



(c)

Fig. 8 Variation of frequency with rotating speed Ω for different power-law indexes g . **a** $m = 1$, $n = 1$, **b** $m = 2$, $n = 1$, **c** $m = 1$, $n = 10$. ($L = 5$ m, $R = 1$ m, $h = 50$ mm, $N_1 = 30$)

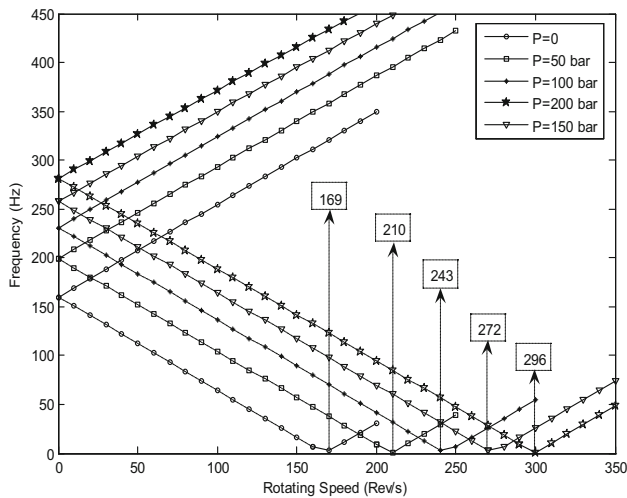


Fig. 9 Sensitivity of critical speed of a rotating FGM cylindrical shell to initial pressure ($L = 5$ m, $R = 1$ m, $h = 6$ mm, $g = 2$, $m = 1$, $n = 1$, $N_1 = 30$)

Appendix

$$K_{11}^1 = \left[C_{11} \left(\frac{m\pi}{L} \right)^2 \alpha_j^2 + C_{55} \left(\frac{\partial \alpha_j}{\partial z} \right)^2 + \frac{C_{66} n^2 \alpha_j^2}{R_j^2} \right] \frac{\pi L}{2} \quad (39)$$

$$K_{12}^1 = \left(-\frac{(C_{66} + C_{12})}{R_j} n \frac{m\pi}{L} \alpha_j^2 \right) \frac{\pi L}{2} \quad (40)$$

$$K_{13}^1 = \left[-(C_{55} + C_{13}) \frac{\partial \alpha_j}{\partial z} \frac{m\pi}{L} \alpha_j - \frac{C_{12} m\pi}{R_j L} \alpha_j^2 \right] \frac{\pi L}{2} \quad (41)$$

$$K_{14}^1 = \left[C_{11} \left(\frac{m\pi}{L} \right)^2 \beta_j \alpha_j + C_{55} \left(\frac{\partial \beta_j}{\partial z} \right) \left(\frac{\partial \alpha_j}{\partial z} \right) + \frac{C_{66} n^2 \beta_j \alpha_j}{R_j^2} \right] \frac{\pi L}{2} \quad (42)$$

$$K_{15}^1 = \left[-(C_{66} + C_{12}) \frac{n}{R_j} \frac{m\pi}{L} \beta_j \alpha_j \right] \frac{\pi L}{2} \quad (43)$$

$$K_{16}^1 = \left[-(C_{55} + C_{13}) \frac{m\pi}{L} \frac{\partial \beta_j}{\partial z} \alpha_j - \frac{C_{12} m\pi}{R_j L} \beta_j \alpha_j \right] \frac{\pi L}{2} \quad (44)$$

$$K_{22}^1 = \left[(C_{22} n^2 + C_{44}) \frac{\alpha_j^2}{R_j^2} + C_{44} \left(\frac{\partial \alpha_j}{\partial z} \right)^2 + C_{66} \left(\frac{m\pi}{L} \right)^2 \alpha_j^2 - \frac{C_{44}}{R_j} \frac{\partial \alpha_j}{\partial z} \alpha_j - \frac{C_{44}}{R_j} \frac{\partial \alpha_j}{\partial z} \alpha_j \right] \frac{\pi L}{2} \quad (45)$$

$$K_{23}^1 = \left[(C_{22} + C_{44}) \frac{n}{R_j^2} \alpha_j^2 + (-C_{44} + C_{23}) \frac{n}{R_j} \frac{\partial \alpha_j}{\partial z} \alpha_j \right] \frac{\pi L}{2} \quad (46)$$

$$K_{24}^1 = \left[-(C_{66} + C_{12}) \frac{n}{R_j} \frac{m\pi}{L} \beta_j \alpha_j \right] \frac{\pi L}{2} \quad (47)$$

$$K_{25}^1 = \left[(C_{22} n^2 + C_{44}) \frac{\beta_j \alpha_j}{R_j^2} + C_{44} \left(\frac{\partial \beta_j}{\partial z} \right) \left(\frac{\partial \alpha_j}{\partial z} \right) + C_{66} \left(\frac{m\pi}{L} \right)^2 \beta_j \alpha_j - \frac{C_{44}}{R_j} \frac{\partial \alpha_j}{\partial z} \beta_j - \frac{C_{44}}{R_j} \frac{\partial \beta_j}{\partial z} \alpha_j \right] \frac{\pi L}{2} \quad (48)$$

$$K_{26}^1 = \left[(C_{22} + C_{44}) \frac{n}{R_j^2} \beta_j \alpha_j + (C_{44} + C_{23}) \frac{n}{R_j} \frac{\partial \beta_j}{\partial z} \alpha_j \right] \frac{\pi L}{2} \quad (49)$$

$$K_{33}^1 = \left[(C_{22} + C_{44} n^2) \frac{\alpha_j^2}{R_j^2} + C_{33} \frac{\partial \alpha_j}{\partial z} \frac{\partial \alpha_j}{\partial z} + C_{55} \left(\frac{m\pi}{L} \right)^2 \alpha_j^2 + \frac{C_{23}}{R_j} \frac{\partial \alpha_j}{\partial z} \alpha_j + \frac{C_{23}}{R_j} \frac{\partial \alpha_j}{\partial z} \alpha_j \right] \frac{\pi L}{2} \quad (50)$$

$$K_{36}^1 = \left[(C_{22} + C_{44} n^2) \frac{\beta_j \alpha_j}{R_j^2} + C_{33} \frac{\partial \beta_j}{\partial z} \frac{\partial \alpha_j}{\partial z} + C_{55} \left(\frac{m\pi}{L} \right)^2 \beta_j \alpha_j + \frac{C_{23}}{R_j} \frac{\partial \alpha_j}{\partial z} \beta_j + \frac{C_{23}}{R_j} \frac{\partial \beta_j}{\partial z} \alpha_j \right] \frac{\pi L}{2} \quad (51)$$

$$K_{42} = \left(-\frac{(C_{66} + C_{12})}{R_j} n \frac{m\pi}{L} \alpha_j \beta_j \right) \frac{\pi L}{2} \quad (52)$$

$$K_{43}^1 = \left[-(C_{55} + C_{13}) \frac{\partial \alpha_j}{\partial z} \frac{m\pi}{L} \beta_j - \frac{C_{12} m\pi}{R_j L} \alpha_j \beta_j \right] \frac{\pi L}{2} \quad (53)$$

$$K_{44}^1 = \left[C_{11} \left(\frac{m\pi}{L} \right)^2 \beta_j^2 + C_{55} \left(\frac{\partial \beta_j}{\partial z} \right)^2 + \frac{C_{66} n^2 \beta_j^2}{R_j^2} \right] \frac{\pi L}{2} \quad (54)$$

$$K_{45} = \left[-(C_{66} + C_{12}) \frac{n}{R_j} \frac{m\pi}{L} \beta_j^2 \right] \frac{\pi L}{2} \quad (55)$$

$$K_{46}^1 = \left[-(C_{55} + C_{13}) \frac{\partial \beta_j}{\partial z} \frac{m\pi}{L} \beta_j - \frac{C_{12} m\pi}{R_j L} \beta_j^2 \right] \frac{\pi L}{2} \quad (56)$$

$$K_{53}^1 = \left[(C_{22} + C_{44}) \frac{n \alpha_j \beta_j}{R_j^2} + (C_{44} + C_{23}) \frac{n}{R_j} \frac{\partial \alpha_j}{\partial z} \beta_j \right] \frac{\pi L}{2} \quad (57)$$

$$K_{55}^1 = \left[(C_{22} n^2 + C_{44}) \frac{\beta_j^2}{R_j^2} + C_{44} \left(\frac{\partial \beta_j}{\partial z} \right)^2 + C_{66} \left(\frac{m\pi}{L} \right)^2 \beta_j^2 - \frac{C_{44}}{R_j} \frac{\partial \beta_j}{\partial z} \beta_j - \frac{C_{44}}{R_j} \frac{\partial \beta_j}{\partial z} \beta_j \right] \frac{\pi L}{2} \quad (58)$$

$$K_{56}^1 = \left[(C_{22} + C_{44}) \frac{n \beta_j^2}{R_j^2} + (-C_{44} + C_{23}) \frac{n}{R_j} \frac{\partial \beta_j}{\partial z} \beta_j \right] \frac{\pi L}{2} \quad (59)$$

$$K_{66}^1 = \left[(C_{22} + C_{44}n^2) \frac{\beta_j^2}{R_j^2} + C_{33} \left(\frac{\partial \beta_j}{\partial Z} \right)^2 + C_{55} \left(\frac{m\pi}{L} \right)^2 \beta_j^2 + \frac{C_{23}}{R_j} \frac{\partial \beta_j}{\partial Z} \beta_j + \frac{C_{23}}{R_j} \frac{\partial \beta_j}{\partial Z} \beta_j \right] \frac{\pi L}{2} \quad (60)$$

$$K_{ij}^1 = K_{ji}^1 \quad (61)$$

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